



Carisurg

Assignment 16

Interim Documentation

Due Date: July 11th, 2026.

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Setup & Data Decisions

1. **Dataset: 55,121 ED encounters from triage_cleaned_v1.csv** — the canonical Week 5 pipeline output.
2. **Random seed:** 42.
3. **Train/test split:**
 - a. 80/20 stratified on ESI, preserving class distribution in both sets.
 - b. **Train:** 44,096 rows
 - c. **Test:** 11,025 rows.
4. **Leakage exclusions (hard rule):** disposition and previous disposition excluded — both are only known after the encounter ends and would leak the answer into the feature set.
5. **Arrival mode encoding:** arrival mode encoded as a binary ambulance flag.
 - a. The value is ‘ambulance’ (lowercase) in this dataset.
 - b. 18,561 ambulance arrivals (33.7%).
6. **Class imbalance:**
 - a. class_weight=‘balanced’ applied to both classifiers.
 - b. ESI 1 = 77 patients (0.14% of 55,121 total).
 - c. Without balancing, the model learns to ignore ESI 1 almost entirely.
7. **Features:** Top-10 shortlist from the Week 5 memo:
 - a. age,
 - b. triage_vital_o2 (SpO₂),
 - c. cc_chestpain,
 - d. cc_shortnessofbreath,
 - e. cc_suicidal,

- f. cc_alcoholintoxication,
- g. cc_alteredmentalstatus,
- h. arrival_ambulance,
- i. triage_vital_rr,
- j. And triage_vital_hr.

Primary Metric Justification

The primary metric is ESI-1 recall, with macro-averaged recall across all five classes used as the summary metric.

ESI 1 patients — cardiac arrest, major trauma, anaphylaxis, shock — require an immediate resuscitation bay response. A false negative on ESI 1 does not mean a slightly delayed response: in cardiac arrest, survival probability drops roughly 10% per minute without intervention. A model that classifies a Level 1 patient as Level 2 or 3 sends them into a waiting queue. The queue is the wrong place for them.

In this dataset, ESI 1 is 0.14% of encounters — 77 patients in total, 16 in the test set. Overall accuracy is therefore clinically meaningless: a model that never predicts ESI 1 at all achieves 99.86% accuracy while failing every Level 1 patient it sees.

Macro vs weighted F1: Weighted F1 is dominated by ESI 3 (49% of data) — a model that ignores ESI 1 entirely can still score a high weighted F1. Macro F1 gives equal weight to every class; ESI-1 performance is not buried. Both are reported, but macro recall on ESI 1 is the primary optimisation target.

Failure mode we are most worried about: false negatives on ESI 1 — a cardiac arrest patient classified as ESI 2 or 3, sent to wait in a general acute cubicle.

Results

Head-to-Head Comparison

Model	Accuracy	Macro Recall	ESI-1 Recall	Weighted F1	Macro F1
Dummy (stratified)	0.375	0.204	~0.14	0.375	0.204
Logistic Regression	0.254	0.417	0.750	0.327	0.214
Decision Tree (depth=5)	0.255	0.386	0.563	0.302	0.170

Logistic regression is the stronger baseline on every metric that matters. ESI-1 recall of 0.750 means it caught 12 of 16 ESI-1 patients in the test set. Random guessing caught approximately 2. Both models beat the dummy on macro recall, confirming they extract genuine signal from the features.

Logistic Regression — Per-Class Results

Class	Precision	Recall	F1-Score	Support
ESI 1	0.00	0.75	0.01	16
ESI 2	0.69	0.31	0.42	3,585
ESI 3	0.58	0.20	0.30	5,402
ESI 4	0.25	0.26	0.26	1,779

ESI 5	0.05	0.57	0.09	243
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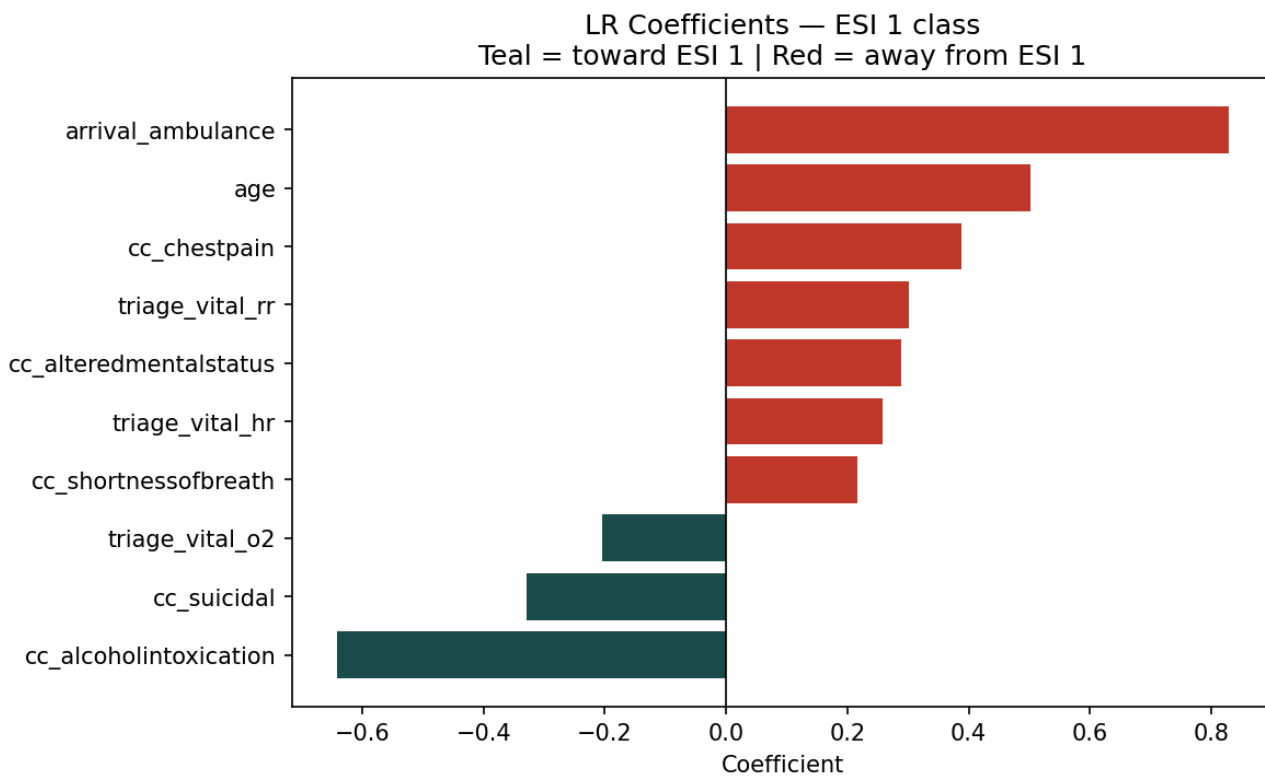
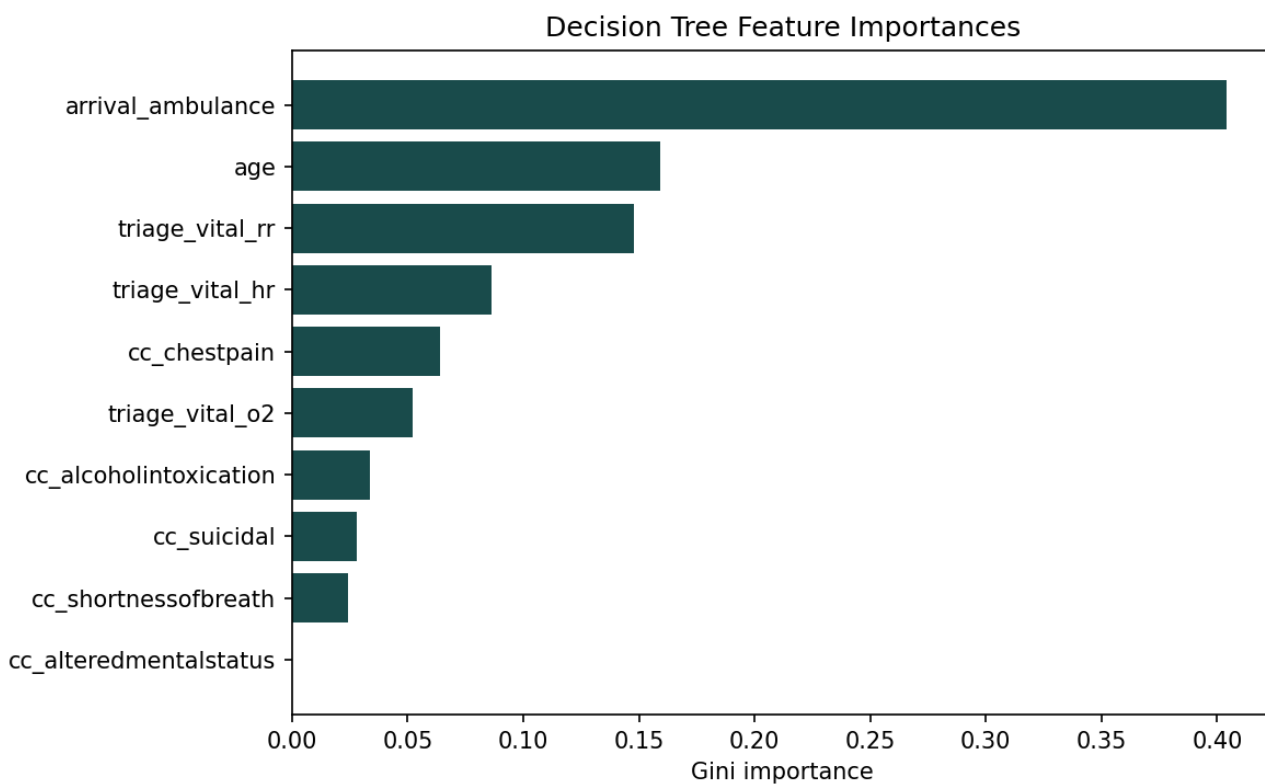
Decision Tree (max_depth=5) — Per-Class Results

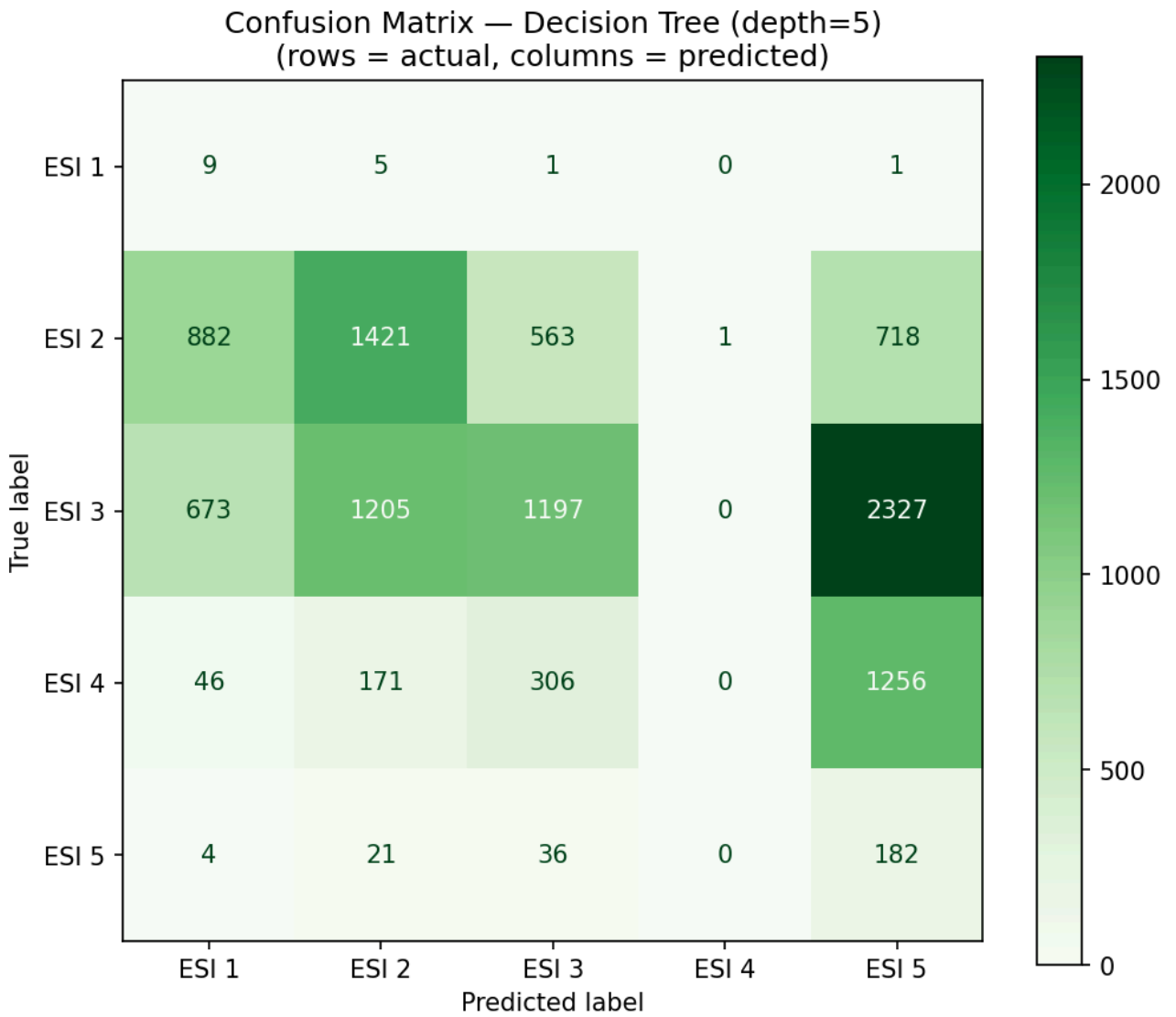
Class	Precision	Recall	F1-Score	Support
ESI 1	0.01	0.56	0.01	16
ESI 2	0.50	0.40	0.44	3,585
ESI 3	0.57	0.22	0.32	5,402
ESI 4	0.00	0.00	0.00	1,779
ESI 5	0.04	0.75	0.08	243

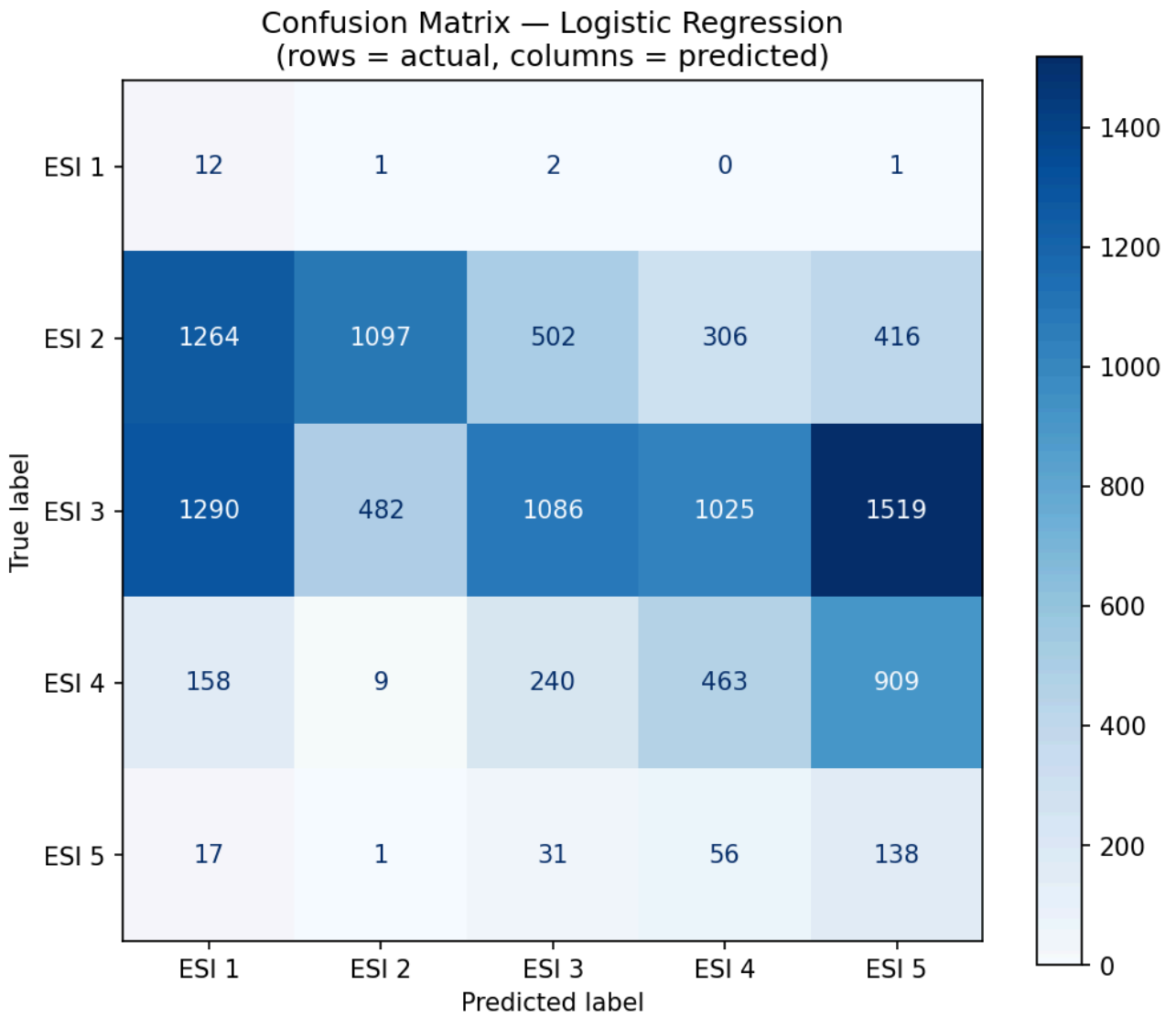
Note: DT assigns zero patients to ESI 4 entirely. This is a known behaviour of shallow decision trees under class-balanced weighting on highly imbalanced data — the tree cannot find a split that improves on the weighted Gini impurity for ESI 4 at depth 5. Worth investigating with a deeper tree or adjusted class weights in future iterations.

LR Coefficient Sanity Check

For the ESI 1 class, a positive coefficient means the feature pushes the prediction toward ESI 1 (more urgent). Clinical expectations: chest pain, altered mental status, suicidal presentation, and ambulance arrival should all be positive; SpO₂ should be negative (lower oxygen = more urgent).







Failure Mode Reflection

1. **Primary concern:** LR misses 4 of 16 ESI-1 patients (25% false negative rate). In a deployed context, those patients join a waiting queue rather than go to the resuscitation bay. For a cardiac arrest patient, this outcome is unsurvivable.
2. **ESI-2 recall failure:** LR recall for ESI 2 = 0.31. ESI-2 patients — stroke, sepsis, ACS, severe pain — have a 10-minute physician target. Classifying them as ESI 1, 4, or 5 at the rates shown means many would reach a physician too late or too early. This is a direct consequence of `class_weight='balanced'`: the algorithm up-weights rare classes at the cost of majority-class precision. This trade-off requires threshold calibration before any deployment.
3. **DT ESI-4 failure:** The decision tree assigns ESI 4 to zero patients. All 1,779 ESI-4 patients (semi-urgent, one-resource) are misclassified. A depth-5 tree cannot represent this class under balanced weighting. A deeper tree or a different weighting scheme is needed.
4. **No re-triage capability:** A static classifier does not flag patients who deteriorate post-triage. A patient correctly classified as ESI 3 at arrival who deteriorates to ESI 1 within 30 minutes is not re-flagged. This is the re-triage constraint from Week 3 — the tool must complement the nurse's visual scanning, not replace it.
5. **ESI-1 sample caveat:** 16 ESI-1 patients in the test set. Each prediction shifts recall by 6.25 percentage points. The 0.750 figure is directionally encouraging but statistically fragile. Do not over-interpret it.
6. **Caribbean mismatch:** The overriding caveat on everything above. This is a US, single-hospital, majority White/Black dataset. Nothing here tests whether these models would transfer to Mercer General. Phase 1 — building Caribbean-sourced data — remains the prerequisite for any deployment-oriented claim.

Macro vs Weighted F1

Weighted F1 weights each class by the number of patients it contains. Because ESI 3 is 49% of the test set, weighted F1 is overwhelmingly determined by ESI-3 performance. A model that classified every patient as ESI 3 would achieve a high weighted F1 while being clinically useless.

Macro F1 gives equal weight to every class. ESI 1 (0.14% of data) counts the same as ESI 3 (49%). A model that fails ESI-1 patients entirely will see this reflected in the macro F1, not hidden under majority-class performance.

For clinical triage decision support, macro recall on ESI 1 is the number to watch. Macro F1 is the summary across all classes. Weighted F1 is reported for completeness but is not the primary optimisation target.

Appendices

Supplementary Links

1. The GitHub link can be found [here](#)
2. Evidence for every figure above is included alongside this memo in the GitHub:
 - a. **figs/01-06_*.png**
 - b. **notebooks/Week6_Baseline_Model**

Glossary

A reference index for clinical, technical, and research terminology used throughout this proposal. Intended for any reviewer — clinical or non-clinical — who needs a quick definition without breaking flow.

Clinical & Vital Sign Terms

Term	Meaning
ED	Emergency Department
ESI	Emergency Severity Index — the 5-level triage scale used at Mercer General (1 = immediate, 5 = non-urgent)
SBP	Systolic Blood Pressure — the peak pressure during a heartbeat (top number in a BP reading)
DBP	Diastolic Blood Pressure — the pressure between heartbeats (bottom number in a BP reading)
MAP	Mean Arterial Pressure — average pressure driving blood to organs across a cardiac cycle; calculated as $(SBP + 2 \times DBP) / 3$
GCS	Glasgow Coma Scale — score from 3–15, measuring level of consciousness
RR	Respiratory Rate — breaths per minute

FiO2	Fraction of Inspired Oxygen — the percentage of oxygen a patient is receiving (room air = 21%)
SpO2	Oxygen Saturation — percentage of haemoglobin carrying oxygen, measured via pulse oximeter
SBAR	Situation, Background, Assessment, Recommendation — a structured format for clinical handover communication

Healthcare Systems & Infrastructure Terms

Term	Meaning
EHR	Electronic Health Record — digital patient record system; Mercer currently has none for triage, hence the Phase 1 proposal
LMIC	Low- and Middle-Income Country — a World Bank income classification used in global health literature
SIDS	Small Island Developing State — a UN classification for small island nations facing shared infrastructure constraints
Boarding time	The gap between a disposition decision (e.g. “admit”) and the patient physically leaving the ED, usually due to no ward bed being available

AI / Machine Learning & Risk Terms

Term	Meaning
AI	Artificial Intelligence
ML	Machine Learning
AUROC / AUC	Area Under the Receiver Operating Characteristic curve — a model performance metric from 0.5 to 1.0; higher is better
XGBoost	Extreme Gradient Boosting — a machine learning algorithm using an ensemble of decision trees, common for structured/tabular data
Rule-based classifier	A decision-making system using explicit if/then logic rather than a trained statistical model
PPV	Positive Predictive Value — the proportion of positive alerts that turn out to be true positives. Low PPV drives alert fatigue.
Automation bias	The tendency for a human user to over-trust or defer to an automated system's output, even when contradicted by their own judgment
Distribution shift	When a model's performance degrades because the data it encounters in deployment differs statistically from the data it was trained on
Human-in-the-loop	A system design principle requiring a human to review, confirm, or be able to override an AI-generated output before it takes effect

Research & Documentation Terms

Term	Meaning
DOI	Digital Object Identifier — a permanent link format for academic papers
APA	American Psychological Association — the citation style used throughout this proposal (7th edition)
Scoping review	A literature review mapping the breadth of existing research, without necessarily assessing study quality in depth
Systematic review	A more rigorous literature review following a defined search and inclusion protocol